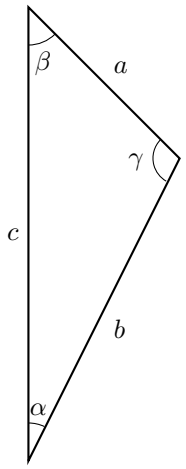
**MATH 195: 10/22**
**RIGHT TRIANGLE TRIGONOMETRY**

Perhaps in high school geometry you learned that three data of angles or sides is usually enough to identify a triangle up to congruence. Using trig you can find the missing info.

**Law of Sines.**

$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$$

This relates two angles and two sides. If you know three, you can solve for the fourth.




**Law of Cosines.**

$$c^2 = a^2 + b^2 - 2ab\cos\gamma,$$

or similarly from the perspective of another angle. This relates three sides and one angle. If you know three, you can solve for the fourth.

**Obtuse angles and inverse trig.**

A triangle can have an angle larger than $\pi/2$ (or $90°$). The circle definitions of the trig functions let us make sense of this. But some care must be taken with inverse trig functions.

- On the top half of the circle, the interval $[0, \pi]$, the cosine function is one-to-one. So it has an inverse restricted to this interval, and your calculator will give the value you want with arccos.
  - If $x > 0$ then $\arccos x$ is in quadrant I, the interval $(0, \pi/2)$.
  - If $x < 0$ then $\arccos x$ is in quadrant II, the interval $(\pi/2, \pi)$.
- The sine function is *not* one-to-one on the top half of the circle. If all you know is that $\sin\alpha = y > 1$ you cannot know if $\alpha$ is in quadrant I or II.
  - If you use your calculator, arcsin will *always* give you the quadrant I angle, not the quadrant II angle.
  - When solving a triangle using the law of sines, this shows up in the ASS ambiguous case.
  - You have to notice that you are in the ambiguous case; you will not get a calculator error to inform you something is sus.
  - If you are in the ambiguous case, you can find the possible quadrant II angle by subtracting the possible quadrant I angle from a half circle ($\pi$ or $180°$).

## Practice problems

For all problems, the best way to start is to draw a picture.

Please give both exact answers in terms of trig functions and use a calculator to give an approximate answer rounded to one digit past the decimal point.

(1) You know a triangle has an angle of $45°$ flanked by two sides of length 5. Find the missing side and angles.
(2) You know a triangle has an angle of $60°$ flanked by sides of length 4 and 12. Find the missing side and angles.
(3) You know that the three sides of a triangle have lengths $4, 6, 8$. Find the angles.
(4) You know that a triangle has sides of length $5, 5, 8$. Find the angles.
(5) You know a triangle has a side of length 4 flanked by angles of measure $35°$ and $55°$. Find the missing angle and sides.
(6) You know a triangle has angles of measure $65°$ and $50°$, with the side between the $50°$ angle and the third angle having length 12. Find the missing angle and sides.
(7) You are a surveyor and you need to measure the distance from one side of a length to the opposite. This is hard to do directly, so you decide to use trig to ease the process. Instead of measuring the distance across the lake, you measure the distance from each end of the lake to a third point off to the side. Because you don't have to travel across water, you can measure that the distances to the third point are 842 meters and 941 meters, with the angle in between these distances being $34°$. Determine the distance across the lake.
(8) You can think of the law of cosines as generalizing the Pythagorean theorem, giving you an error term to correct for the length of the "hypotenuse" when $\gamma$ isn't a right angle. Think of $a$ and $b$ as fixed lengths joined at the angle $\gamma$. As you change $\gamma$ the length of the third side $c$ changes.

    Geometrically, it should be clear that $c$ gets smaller as $\gamma$ gets smaller and $c$ gets larger as $\gamma$ gets larger. Draw a picture to illustrate this. Then give an explanation for this fact based on the values of cos in the top half of the unit circle. What values are $c$ approaching as $\gamma$ approaches 0 and as $\gamma$ approaches $\pi$ (or $180°$)?
(9) Explain why the laws of sines and cosines don't let you solve for any side lengths if you only know the three angles of the triangle. Give a geometric argument for why this AAA case doesn't uniquely specify a triangle up to congruence.
(10) sin isn't one-to-one on the top half of the circle, because angles in quadrants I and II will have the same value under sin. But there is one value $y > 0$ so that if $\sin\alpha = y$ then you know there's only one possible value for $\alpha$ in the interval $[0, \pi]$. What is this value $y$ and corresponding angle? Explain.
(11) Sometimes the ASS case isn't ambiguous. This happens, for instance, when the known angle is right. Explain in this case how you would determine the missing side and angles. Give an example and use your procedure to solve for the missing data.
(12) Sometimes the ASS case is ambiguous. This happens when the known angle is acute and the second side is smaller than the first. Suppose you know that a triangle has an angle of $30°$ followed by sides of length 10 and 6. Draw a picture to demonstrate why there are two possible triangles this could be. Then determine the sides and angles for both possible triangles.